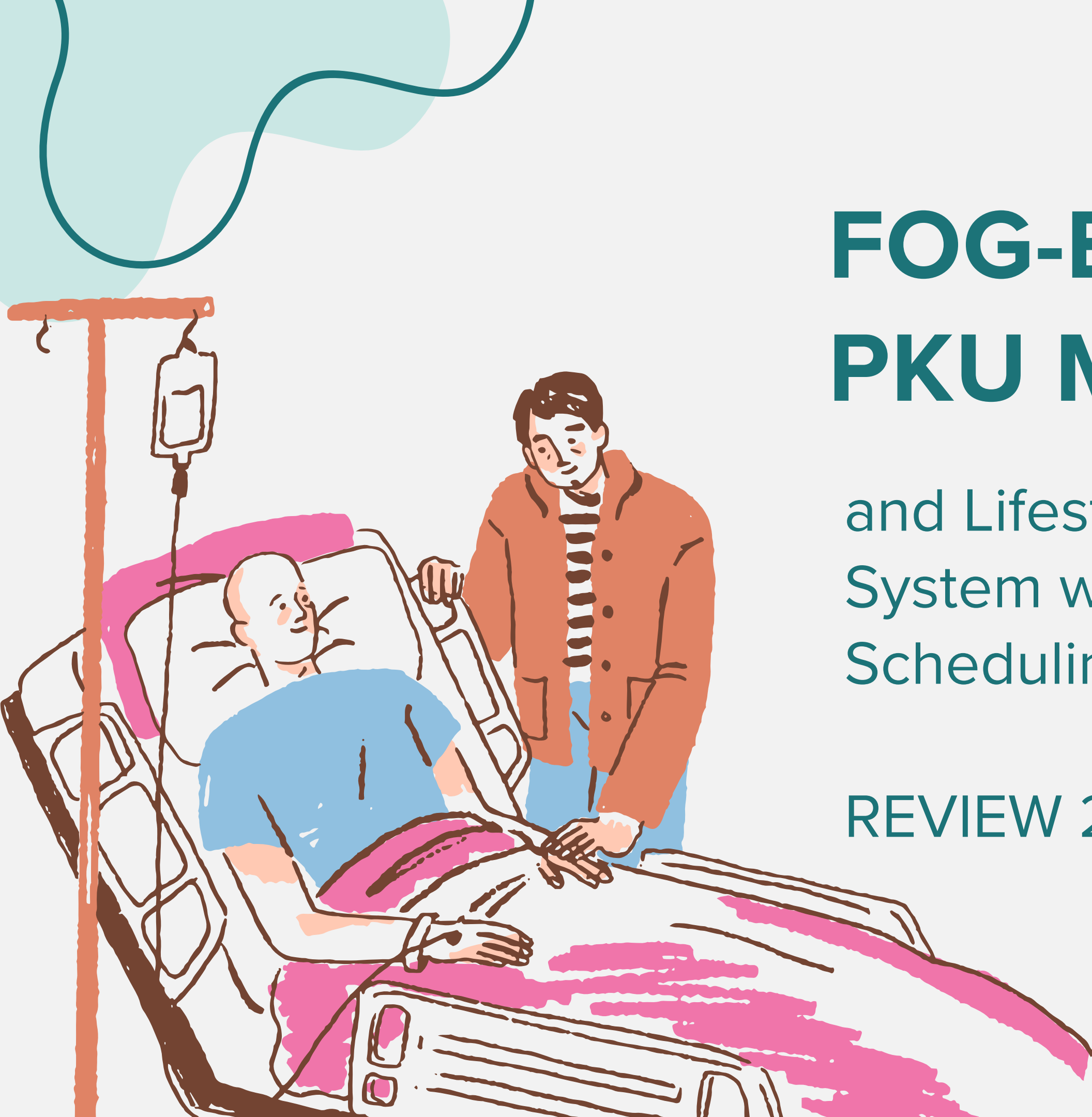




FOG-ENABLED SMART PKU MONITORING

and Lifestyle Recommendation
System with Priority-Based Task
Scheduling

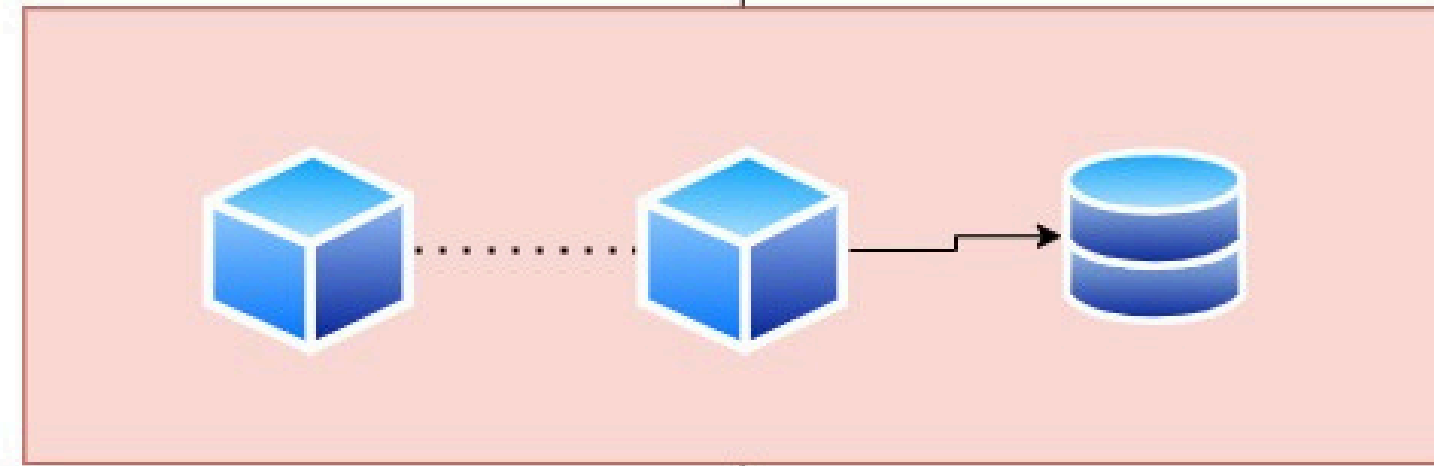
REVIEW 2

CLOUD LAYER



CLOUD PROCESSING
Storage of medical data

FOG LAYER



FOG PROCESSING

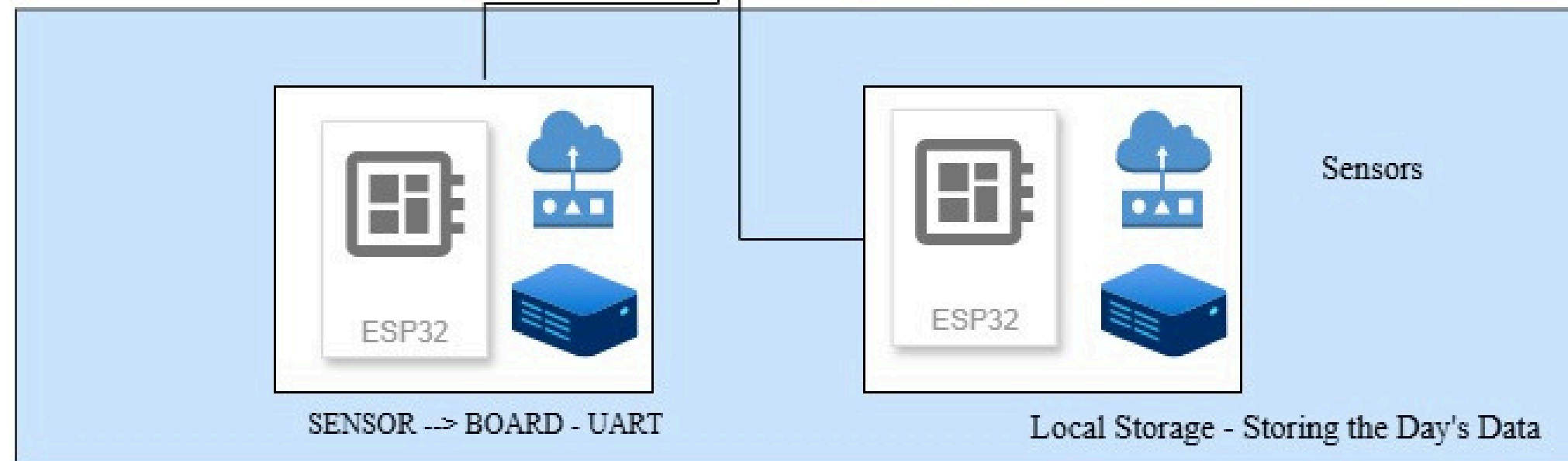
1. ML Inference - Send the dietary suggestions to patients
2. Standby node for extreme cases
3. Local storage for saving periodic data.

EDGE NODE --> FOG NODE - MQTT



Gateway

EDGE LAYER



Sensors

EDGE PROCESSNG

1. Moderate spike - Sent to the fog node for ML inference
2. Crtical Spike - Sent to patient via LoRa and fog node .
3. Battery low Alert - Sent to fog node
4. Store the day's data and send it at night for ML inference

FUZZY RULES

How It Works

- Processes Phenylalanine (Phe) & Tyrosine (Tyr) sensor values.
- Maps values into fuzzy sets: Normal, Slightly High, Dangerous.
- Example Rules:
 - IF Phe Normal AND Tyr Normal → Safe
 - IF Phe Slightly High OR Tyr Slightly High → Borderline
 - IF Phe High AND Tyr Low → Dangerous
- Produces confidence levels (e.g., 70% Dangerous, 30% Borderline).

Why Fuzzy Rules

- Manage sensor uncertainty (noisy biosensor data).
- Capture gradual clinical changes (not abrupt thresholds).
- Lightweight → suitable for edge devices.
- Provides real-time alerts before NN/PSO processing.

Results

- Safe: No action.
- Borderline: Forward data to fog for NN prediction.
- Dangerous: Immediate alert to patient/clinician.
- Output includes confidence score for better interpretability.

NEURAL NETWORKS

How It Works

- Runs at the fog layer for deeper analysis.
- Inputs: Phe, Tyr, hours since last meal, diet logs, patient profile.
- Multi-layer NN learns nonlinear patterns in patient data.
- Outputs: Risk score (probability of Phe spike).

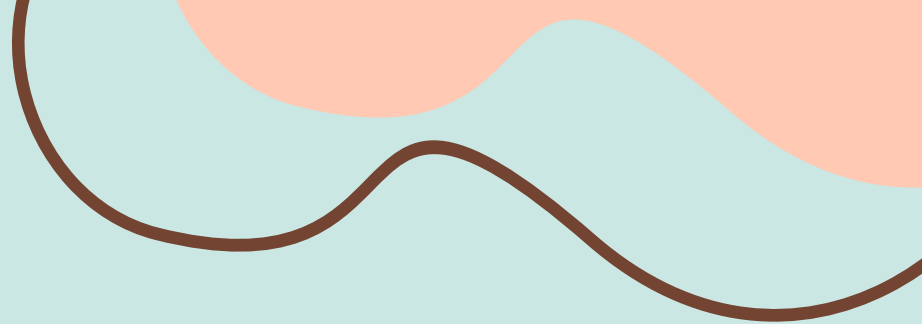
Why Neural Networks

- Handle complex, nonlinear relationships between diet and biochemistry.
- Enable personalized predictions using patient history.
- Go beyond fuzzy rules → add foresight, not just current state.
- Provide proactive insights for PSO diet optimization.

Results

- Risk score: e.g., 0.8 → high probability of spike.
- Trajectory forecast: future Phe/Tyr trends.
- Decision support:
 - High risk → alert clinician.
 - Moderate → adjust diet plan via PSO.
 - Low → routine monitoring.

PSO



How we integrate

- We embed Particle Swarm Optimization (PSO) at the Fog layer.
- Inputs: sensor data (Phe levels, calories consumed) + patient dietary history.
- Each particle represents a possible diet plan (food servings).
- Iteratively updates solutions to minimize phenylalanine while meeting calorie needs.

Why we integrate

- Diet planning is a complex optimization problem with multiple constraints.
- PSO reduces trial-and-error in clinical diet design.
- Provides fast, scalable, and personalized recommendations.

Optimized Results after integration

- Generates personalized diet plans within Phe & calorie limits.
- Suggests clear food adjustments (e.g., increase fruits, reduce rice).
- Produces an optimized recommendation list that can be directly used by patients and clinicians.

WORKFLOW

1. Initialization

- Each particle = a possible diet (quantities of rice, milk, fruits, etc.).

2. Fitness Evaluation

- Fitness = total Phe intake (to minimize) + calorie requirement (to satisfy).

3. Update Bests

- Each particle updates its personal best (pBest).
- Swarm updates global best (gBest).

4. Velocity & Position Update

- Adjust diet quantities toward better solutions.

5. Iteration

- Repeat until optimal diet plan emerges.

Optimized Output

- “Take more fruits, reduce rice, avoid milk”
- Personalized plan aligned with medical constraints.



ALGORITHMS USED

- **EDGE LAYER ALGORITHM - PRIORITY-BASED TASK SCHEDULING**
 - HANDLES MULTIPLE TYPES OF TASKS
- **FOG LAYER ALGORITHM – HEARTBEAT ALGORITHM**
 - CONTINUOUSLY MONITORS HEALTH OF THE PRIMARY FOG NODE.
 - IF THE PRIMARY FOG NODE BECOMES UNRESPONSIVE, THE STANDBY NODE TAKES OVER PROCESSING
 - PROVIDES FAULT TOLERANCE AND SYSTEM RELIABILITY.

CONCLUSION

This project presents a real-time healthcare monitoring system using edge and fog computing. By simulating a sensor patch we ensured realistic input conditions. The smartphone acts as the edge device, the fog computing layer comprising a main hospital node and backup nodes that handles task scheduling, deeper ML analysis, alerts and daily reports. This end-to-end setup ensures fast, reliable, and scalable health monitoring system.

